

# Markets and Coordination

Principles of Microeconomics

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# Millions of People Plan the Groceries. What Coordinates Their Plans?

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**Millions of strangers grow, ship, and sell the groceries others want, in roughly the right amounts and places.**

Bakers choose how much bread to bake.  
Retailers choose how much milk to order. No single planner directs all their choices.

**What connects their plans? And when the shelves do go empty, what went wrong?**



## A Middleman Can Raise the Cash Price and Lower the Full Cost

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A used car is listed at \$5,000. The weekends of searching, the inspection, and the risk of being cheated are **transaction costs**; cash plus those is the **full price**. A **middleman** buys to resell and does that work for a fee.

| One egg            | Money | Nonmoney          | Buyer's full cost / seller's net receipt |
|--------------------|-------|-------------------|------------------------------------------|
| Buyer, on his own  | 8.0¢  | 1.0¢ of trouble   | 9.0¢ paid                                |
| Seller, on his own | 8.0¢  | 0.5¢ selling cost | 7.5¢ received                            |

**Buyer and seller together bear 1.5¢ in transaction costs.** A middleman offers to do the work: more cash from the buyer, less to the seller. Is either side worse off?

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| Buyer, via a middleman  | 8.5¢  | 0                 | <b>8.5¢</b> paid                         |
| Seller, via a middleman | 7.75¢ | 0                 | <b>7.75¢</b> received                    |

The buyer's cash price rises to 8.5¢; his full price falls from 9.0 to 8.5. The seller's receipt falls to 7.75¢; his net rises from 7.5 to 7.75.

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**Buyer saves 0.5¢; seller gains 0.25¢; middleman's spread is 0.75¢.** The spread replaces 1.5¢ of DIY cost; profit is the spread less his own costs.

## Who Gains When the City Caps Taxi Licenses?

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**Open market:** anyone may enter. **Closed:** entry restricted by force, collusion, or law.

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## **INSIDERS**

### **A protected margin**

Fewer rivals, higher fares.

The license becomes  
valuable.

## **THE EXCLUDED**

### **Blocked trades**

Riders pay more; some  
rides never happen. Losses  
spread thin.

## **THE LOBBYING**

### **Rent-seeking**

Real resources spent to win  
a favor, not to produce a  
ride.

**Freedom of competition is open entry. Freedom from it is protection.**

The margin is a transfer; lost rides and lobbying add costs. Screening benefits must be demonstrated.

# The Price of Eggs Doubles Overnight. What Does That One Number Do?

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## **SIGNAL**

**What does the baker learn?**

## **REWARD**

**What changes in orders and farm plans?**

## **RATIONS**

**Who buys the last dozen?**

# The Price of Eggs Doubles Overnight. What Does That One Number Do?

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## **SIGNAL**

**Sums up demands and supply**

A pileup lowers it;  
competition for what exists  
raises it.

## **REWARD**

**It restrains buyers and  
pays suppliers**

When output can respond,  
sellers have a reason to  
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**Whoever will give up  
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## **People revise their plans in response to prices.**

One seller's price is the next producer's cost: cattle, butcher, steak. Under enforceable rights, workable competition, usable information, and room to adjust.

## Add Every Buyer's Demand at Each Price. Seven Cars Exist. Where Do They Clear?

---

| Price    | A wants | B | C | D |
|----------|---------|---|---|---|
| \$10,000 | 1       | 0 | 1 | 1 |
| \$9,000  | 1       | 1 | 1 | 2 |
| \$8,000  | 2       | 1 | 1 | 2 |
| \$7,000  | 2       | 1 | 2 | 2 |
| \$6,000  | 2       | 2 | 2 | 2 |
| \$5,000  | 2       | 2 | 2 | 2 |
| \$4,000  | 2       | 2 | 2 | 3 |
| \$3,000  | 3       | 2 | 2 | 3 |
| \$2,000  | 3       | 3 | 2 | 3 |
| \$1,000  | 4       | 4 | 2 | 4 |

**FOUR BUYERS, ONE STOCK**  
**Each wants some number  
of cars at each price**

Seven cars exist, all alike. None  
can be built. Which price makes  
the four want exactly seven?

**Count the cars the four want at each price.** Where does the total equal seven?

## Add Every Buyer's Demand at Each Price. Seven Cars Exist. Where Do They Clear?

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| Price    | A wants | B | C | D | Market |
|----------|---------|---|---|---|--------|
| \$10,000 | 1       | 0 | 1 | 1 | 3      |
| \$9,000  | 1       | 1 | 1 | 2 | 5      |
| \$8,000  | 2       | 1 | 1 | 2 | 6      |
| \$7,000  | 2       | 1 | 2 | 2 | 7      |
| \$6,000  | 2       | 2 | 2 | 2 | 8      |
| \$5,000  | 2       | 2 | 2 | 2 | 8      |
| \$4,000  | 2       | 2 | 2 | 3 | 9      |
| \$3,000  | 3       | 2 | 2 | 3 | 10     |
| \$2,000  | 3       | 3 | 2 | 3 | 11     |
| \$1,000  | 4       | 4 | 2 | 4 | 14     |

### MARKET DEMAND

**The sum of the four, at each price**

Add across a row: at \$8,000 the market wants 6; at \$6,000, 8.

**Market demand is the sum of the individual demands at each price.**

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|----------------|----------|----------|----------|----------|-------------------|
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| \$9,000        | 1        | 1        | 1        | 2        | 5                 |
| \$8,000        | 2        | 1        | 1        | 2        | 6                 |
| <b>\$7,000</b> | <b>2</b> | <b>1</b> | <b>2</b> | <b>2</b> | <b>7 = 7 cars</b> |
| \$6,000        | 2        | 2        | 2        | 2        | 8                 |
| \$5,000        | 2        | 2        | 2        | 2        | 8                 |
| \$4,000        | 2        | 2        | 2        | 3        | 9                 |
| \$3,000        | 3        | 2        | 2        | 3        | 10                |
| \$2,000        | 3        | 3        | 2        | 3        | 11                |
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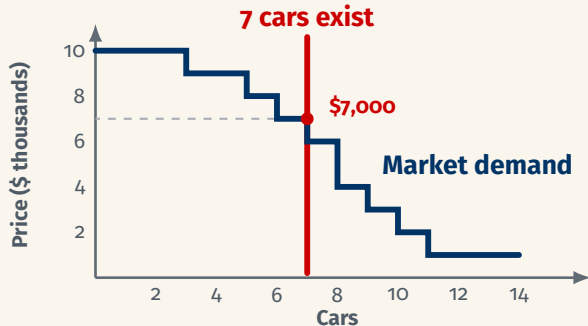
### CLEARING PRICE

**\$7,000: seven wanted,  
seven exist**

Above it a car goes unsold; below it buyers bid up. With fixed demands and low trading costs, initial ownership does not change this row.

**The market clears at \$7,000:** amount demanded equals amount available.

# A Rising Price Ratios Even a Stock That Cannot Grow



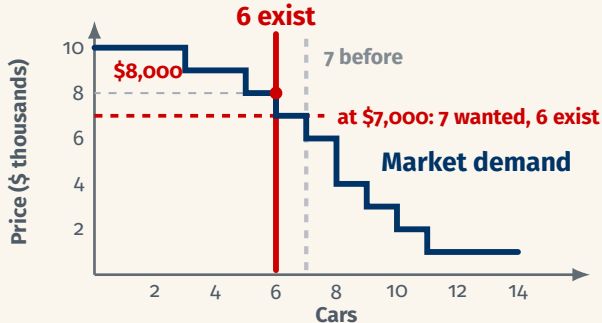
## THE PICTURE

**The staircase is the table's Market column**

The red line is the seven cars that exist. They meet at \$7,000.

**Seven cars, no factory.** Can a higher price do any good if not one more car can be made?

# A Rising Price Ratios Even a Stock That Cannot Grow



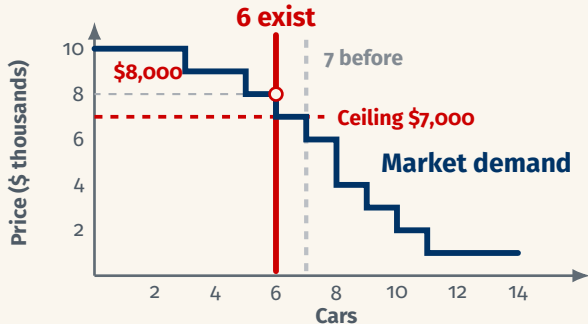
## A'S CAR BURNS

**A has one car left. At \$8,000 he wants two.**

C wants only one at \$8,000. A buys C's second car. Both gain from this trade; demands stay fixed.

**A buys from C at \$8,000: A 2, B 1, C 1, D 2.** C gains from the sale; A is better off than after the fire without trading.

# A Rising Price Ratios Even a Stock That Cannot Grow



**A CEILING AT \$7,000**

**No one sells at that price. A cannot replace his car.**

Owners keep the six surviving cars: A 1, B 1, C 2, D 2. Seven are wanted at the ceiling.

**Impose the ceiling before the replacement trade.** Existing ownership determines who keeps the cars; the mutually beneficial sale is blocked.

# A Shortage Is Not Scarcity, and a Surplus Is Not Abundance

---

| Word     | What it is | What makes it | In the car market |
|----------|------------|---------------|-------------------|
| Scarcity | ?          | ?             | ?                 |

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| Word                       | What it is                                      | What makes it         | In the car market                    |
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| <b>Scarcity</b>            | Less exists than people want at a price of zero | The human condition   | Seven cars, many wants               |
| <b>Reduction in supply</b> | The amount that exists falls                    | A fire, a failed crop | Six cars, price \$8,000: no shortage |

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**Here the held price makes the shortage, not the stock's size.**

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| <b>Surplus</b>             | More offered than bought; the gap persists      | A price held above clearing | Reset: 7 cars. At \$8,000: 6 wanted, 1 unsold               |

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# Three Ideas Explain How Strangers Get Coordinated

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1. A middleman raises the cash price of an egg. Why can the buyer still be better off?
2. Taxi licenses are capped. Name the winners, the losers, and the word for the lobbying.
3. A fire cuts the stock to six and the price is free to rise. Is there a shortage? Which word applies instead?

**NEXT: PRICE CONTROLS**    **A law forbids the price to move. What moves instead?**

# Three Ideas Explain How Strangers Get Coordinated

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## FULL PRICE

**Money price plus  
every nonmoney cost**

A middleman can raise the cash price and still leave both sides better off.

## OPEN ENTRY

**Closing a market  
transfers wealth**

A protected margin for insiders; blocked trades for everyone else.

## CLEARING PRICE

**Amount wanted equals  
what exists**

Signals, rewards, rations. A shortage is the gap a held price leaves.

**NEXT: PRICE CONTROLS  
moves instead?**

**A law forbids the price to move. What**

# A

# Applications

Each one stands on its own. Same three ideas, new places.

## **Ten Cartons Are for Sale. Twenty Shoppers Want One. Who Gets Them?**

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**A store offers 10 egg cartons at a capped price. Twenty shoppers each want one. The store will sell all ten. How could it choose?**

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## **QUEUES**

**First come, first served**

Time in line is a cost, paid to no one.

## **DISCRETION**

**Whoever the seller favors**

Friends, regulars, the official's pick.

## **RULES AND LOTTERIES**

**Eligibility lists, chance**

A rule can favor a target group. It must be enforced and can be gamed.

**Every system rations. Compare the rules, not the labels.**

By information, incentives, distribution, cost, and how errors get corrected.

## Give A All Seven Cars. Where Do They End Up?

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**Start over with A owning all seven cars and the same four demands. Where do the cars end up, and at what price?**

Reference: at \$7,000, desired holdings are A 2   B 1   C 2   D 2.

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## **EACH TRADER'S RULE**

**Buy below your worth;  
sell above it**

Every trade moves a car to someone who values it more.

## **WHERE IT STOPS**

**A 2, B 1, C 2, D 2**

At \$7,000 each holds what they want. No further mutually desired trade remains.

## **WHAT THAT ASSUMES**

**Fixed demands; low trading costs**

Informed, voluntary, enforceable trades; no wealth effect on demand.

**Under these assumptions, initial ownership does not change final holdings.**

## **“Demand Will Outstrip Supply by 2040.” What Is Missing from the Chart?**

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## **THE HIDDEN ASSUMPTION**

### **The price sits still**

Both lines were drawn at today's price.

## **A FREE PRICE**

### **Rises as the good tightens**

Buyers use less; suppliers seek more. Adjustment takes time.

## **THE FIRST QUESTION**

### **Did demand shift, or the amount available?**

Not “is this gouging?” A price change is news first.

**Flexible prices create pressure to close the projected gap.**

## **The Rule Says “Consumer Safety.” How Would You Know?**

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## LOOKS LIKE SCREENING

**Fewer bad outcomes,  
better information**

Buyers who could not judge  
quality now can.

## LOOKS LIKE PROTECTION

**Capable rivals kept  
out; prices above  
comparable services**

The incumbents set the bar.

## THE TEST

**Marginal benefit  
against the costs**

Entry, compliance,  
enforcement, political. Both  
effects can coexist.

**Read the effects, not the label.**

## **Aldi's Pear Is Cheaper. Is Its Full Price Lower?**

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**Aldi charges less for the pear than Publix. You bag your own, dig through the bin, and bring a quarter for the cart.**

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### **MONEY PRICE**

**Lower**

Fewer services on the shelf;  
a smaller spread.

### **NONMONEY COST**

**Higher**

Your time sorting, bagging,  
and carrying. The work  
moved to you.

### **FULL PRICE**

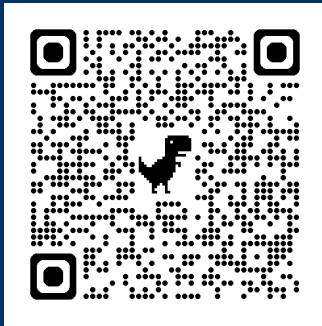
**Depends on what your  
time is worth**

Cheaper for some shoppers,  
dearer for others.

**Both stores are middlemen. They sell different amounts of the work.**

BEFORE YOU GO

# Tell me what worked and what didn't.



Scan for a one-minute anonymous form.